

Skill9

Task1: To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs.

Code:

```
#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>


int main() {

    int fd = open("input.txt", O_RDONLY);


    if (fd < 0) {

        perror("input.txt");

        return 1;

    }


    dup2(fd, STDIN_FILENO);

    close(fd);


    char buf[100];


    ssize_t n = read(STDIN_FILENO, buf, sizeof(buf) - 1);


    if (n < 0) {

        perror("read");

        return 1;

    }

}
```

```

    }

    buf[n] = '\0';

    printf("Input from redirected file:\n%s", buf);

    return 0;
}

```

OUTPUT:

```

ashwithreddy_earla@AshwithReddy:~$ gcc Skill9.c -o Skill9
ashwithreddy_earla@AshwithReddy:~$ ./Skill9
Input from redirected file:
ashwithreddy_earla@AshwithReddy:~$ echo "Hello from input.txt" > input.txt
ashwithreddy_earla@AshwithReddy:~$ ./Skill9
Input from redirected file:
Hello from input.txt
ashwithreddy_earla@AshwithReddy:~$ |

```

Task2 :To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection.

Code:

```

#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

```

```

int main() {

    int fd = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {

        perror("output.txt");

        return 1;
    }
}

```

```

}

dup2(fd, STDOUT_FILENO);

close(fd);

printf("This output is redirected to a file.\n");
printf("Output redirection completed.\n");

return 0;
}

```

OUTPUT:

```

ashwithreddy_earla@AshwithReddy:~$ nano Skill9.1.c
ashwithreddy_earla@AshwithReddy:~$ gcc Skill9.1.c -o Skill9.1
ashwithreddy_earla@AshwithReddy:~$ ./Skill9.1
ashwithreddy_earla@AshwithReddy:~$
ashwithreddy_earla@AshwithReddy:~$ cat output.txt
This output is redirected to a file.
Output redirection completed.

```

Task 3 :To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios.

Code:

```

#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

int main() {

    int fd = open("append.txt", O_WRONLY | O_CREAT | O_APPEND, 0644);

    if (fd < 0) {

```

```

        perror("append.txt");

        return 1;
    }

    dup2(fd, STDOUT_FILENO);

    close(fd);

    printf("New line appended to the file.\n");

    return 0;
}

```

Output:

```

ashwithreddy_earla@AshwithReddy:~$ nano Skill9.2.c
ashwithreddy_earla@AshwithReddy:~$ gcc Skill9.2.c -o Skill9.2
ashwithreddy_earla@AshwithReddy:~$ ./Skill9.2
ashwithreddy_earla@AshwithReddy:~$ cat append.txt
New line appended to the file.
ashwithreddy_earla@AshwithReddy:~$ |

```

Task4: To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases.

Code:

```

#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

int main() {

    int fd = open("error.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

```

```
if (fd < 0)

    return 1;


dup2(fd, STDERR_FILENO);

close(fd);


fprintf(stderr, "Error: file could not be opened.\n");

fprintf(stderr, "stderr redirection completed.\n");


return 0;

}
```

Output:

```
ashwithreddy_earla@AshwithReddy:~$ nano Skill9.3.c
ashwithreddy_earla@AshwithReddy:~$ gcc Skill9.3.c -o Skill9.3
ashwithreddy_earla@AshwithReddy:~$ ./Skill9.3
ashwithreddy_earla@AshwithReddy:~$ cat error.txt
Error: file could not be opened.
stderr redirection completed.
ashwithreddy_earla@AshwithReddy:~$
```